

TrendWatch AI — Social Media Intelligence Report

Use Case 2: Brand monitoring on Reddit · Generated 2026-06-09 03:50 UTC

Query: dog r/animal Limit: 46 Time: devvit_import

Data: Live Reddit (imported from Devvit app)

Report contents (assignment sections)

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1. Executive briefing

Source: rule_based_fallback · RAG + CoT LLM or rule-based fallback

Executive summary

TrendWatch AI analysed 46 Reddit posts. Negative sentiment is 0.0%. Crisis score is 0/100 (Low).

Main concerns

- Discussion appears within normal sentiment bounds.

Crisis explanation

Discussion appears within normal sentiment bounds.

Recommended actions

- Monitor high-impact negative threads daily.
- Prepare a factual FAQ for top complaint topics.
- Escalate to customer support if refund/delay keywords spike.

Communication strategy

Address emerging topics: When / This / Years, This / These / Just, Dogs / Fish / From. Use empathetic, transparent tone; avoid dismissive responses.

Full LLM report (markdown)

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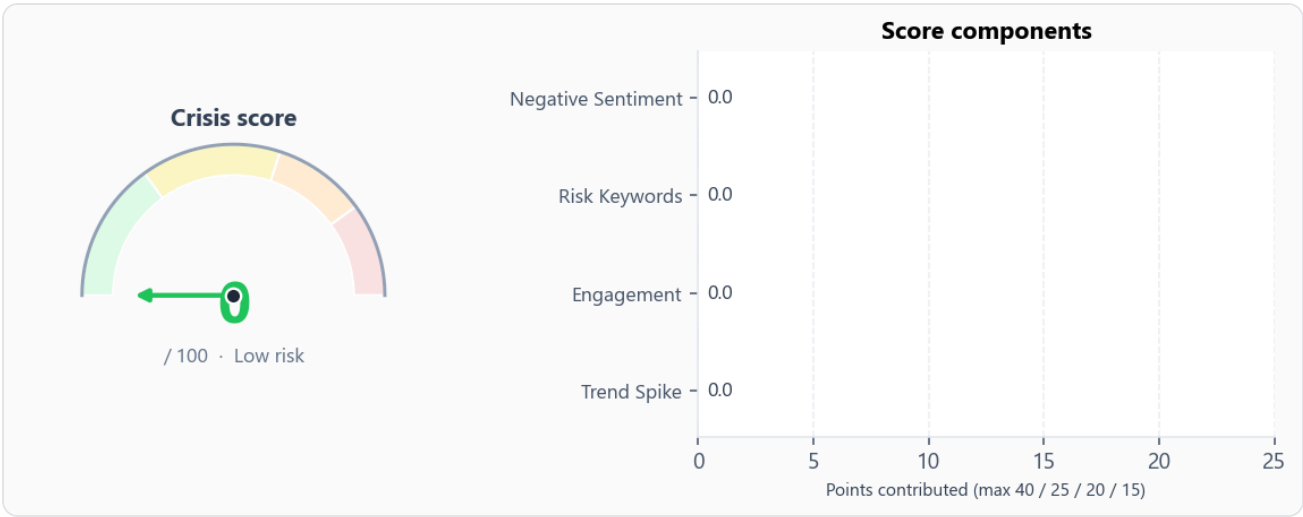
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Communication Strategy

Address emerging topics: When / This / Years, This / These / Just, Dogs / Fish / From. Use empathetic, transparent tone; avoid dismissive responses.

Ethics notice: Analysis uses public Reddit text. Sentiment and crisis scores are algorithmic estimates, not legal or PR advice. Models may reflect language bias; human review is required before brand actions.

2. Crisis intelligence layer



0/100 — Low risk

Weighted fusion of negative sentiment, risk keywords, high-engagement negative posts, and trend spikes.

Score components

Component	Points
negative_sentiment	0.0
risk_keywords	0.0
engagement	0.0
trend_spike	0.0

Reasons

- Discussion appears within normal sentiment bounds.

3. Overview & sentiment analysis

46

POSTS ANALYSED

56.5%

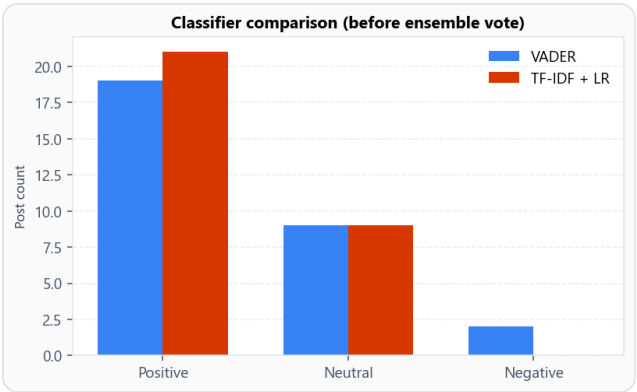
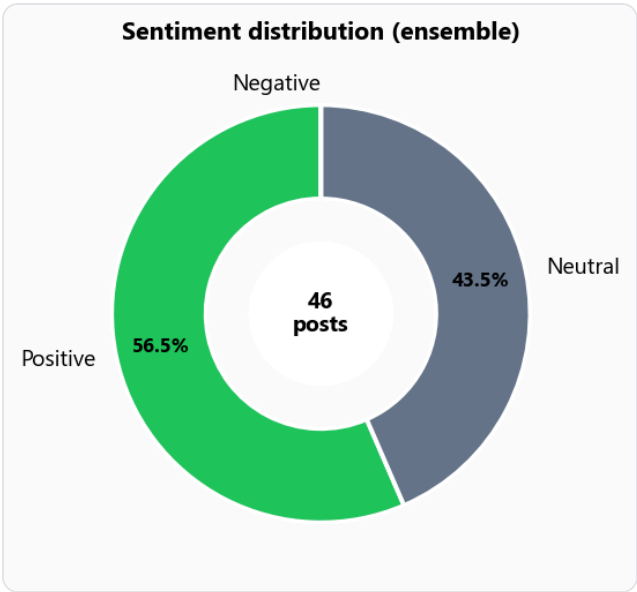
POSITIVE

43.5%

NEUTRAL

0.0%

NEGATIVE



Independent classifier outputs before ensemble fusion.

Final labels after VADER + LR ensemble majority vote.

Sentiment distribution (VADER + LR ensemble)

Positive

56.5%

Neutral

43.5%

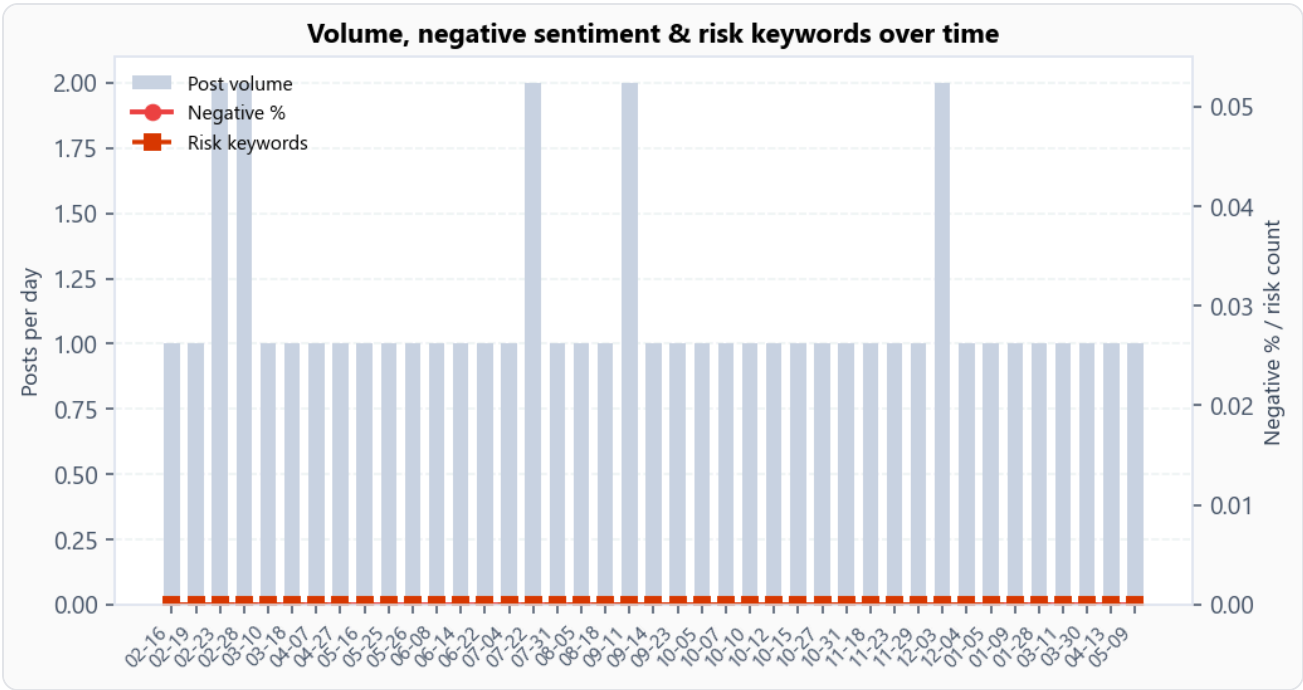
Negative

0.0%

Label	Count
positive	26
neutral	20
negative	0

Counts: positive=26, neutral=20, negative=0

4. Trend detection



Daily post volume (bars), negative sentiment % (line), and risk keyword counts (dashed).

Spike detected: No — No significant negative sentiment spike detected.

Average negative ratio: 0.0%

Date	Posts	Negative	Neg %	Risk keywords
2025-02-16	1	0	0.0%	0
2025-02-19	1	0	0.0%	0
2025-02-23	2	0	0.0%	0
2025-02-28	2	0	0.0%	0
2025-03-10	1	0	0.0%	0
2025-03-18	1	0	0.0%	0
2025-04-07	1	0	0.0%	0
2025-04-27	1	0	0.0%	0
2025-05-16	1	0	0.0%	0
2025-05-25	1	0	0.0%	0

Date	Posts	Negative	Neg %	Risk keywords
2025-05-26	1	0	0.0%	0
2025-06-08	1	0	0.0%	0
2025-06-14	1	0	0.0%	0
2025-06-22	1	0	0.0%	0
2025-07-04	1	0	0.0%	0
2025-07-22	2	0	0.0%	0
2025-07-31	1	0	0.0%	0
2025-08-05	1	0	0.0%	0
2025-08-18	1	0	0.0%	0
2025-09-11	2	0	0.0%	0
2025-09-14	1	0	0.0%	0
2025-09-23	1	0	0.0%	0
2025-10-05	1	0	0.0%	0
2025-10-07	1	0	0.0%	0
2025-10-10	1	0	0.0%	0
2025-10-12	1	0	0.0%	0
2025-10-15	1	0	0.0%	0
2025-10-27	1	0	0.0%	0
2025-10-31	1	0	0.0%	0
2025-11-18	1	0	0.0%	0
2025-11-23	1	0	0.0%	0
2025-11-29	1	0	0.0%	0
2025-12-03	2	0	0.0%	0
2025-12-04	1	0	0.0%	0
2026-01-05	1	0	0.0%	0

Date	Posts	Negative	Neg %	Risk keywords
2026-01-09	1	0	0.0%	0
2026-01-28	1	0	0.0%	0
2026-03-11	1	0	0.0%	0
2026-03-30	1	0	0.0%	0
2026-04-13	1	0	0.0%	0
2026-05-09	1	0	0.0%	0

5. Topic clusters (embeddings + KMeans)

Topic 3: When / This / Years

Size: 16 · Avg sentiment: **neutral**

Keywords: when, this, years, with, loves, like

Representative: i got to meet two capys + a prairie dog

Topic 4: This / These / Just

Size: 9 · Avg sentiment: **positive**

Keywords: this, these, just, know, take, still

Representative: That doesn't sound like a dog..... nobody will kill you....

Topic 0: Dogs / Fish / From

Size: 8 · Avg sentiment: **neutral**

Keywords: dogs, fish, from, your, cats, mushrooms

Representative: What warned my friend

Topic 2: Reason / They / Cool

Size: 8 · Avg sentiment: **positive**

Keywords: reason, they, cool, entered, look, starla

Representative: Looks like we have a home invader (possum)

Topic 1: Produce / Fall / Under

Size: 5 · Avg sentiment: **neutral**

Keywords: produce, fall, under, family, working, will

Representative: What makes a zorse, a zonkey, and a mule any different from a dog mutt?

6. Named entities (NER)

Entity	Mentions
two (CARDINAL)	2
london (GPE)	2
this morning (TIME)	2
doesn (PERSON)	2
3 month old yorkie (DATE)	1
google (ORG)	1
every 24 hours (TIME)	1
margaret school (ORG)	1
gore (PERSON)	1
third (ORDINAL)	1
uk news express co (ORG)	1
an afternoon (TIME)	1
a few hours (TIME)	1
morning (TIME)	1
10 hours (TIME)	1

7. Risk keywords & high-impact posts

Risk keyword frequency

No data.

Similar post pairs (cosine similarity)

Post A	Post B	Similarity
Calling All Dog Lovers 🙏	t3_1je1p2p	0.622
A picture of my dog with a painting of my dog	Photographer session with dogs	0.612
Calling All Dog Lovers 🙏	Recent video of me and my dog of 17 years who died this	0.612
Vote for Starla as Cutest Dog! ✨💖	These are my own little puppies who entered a pet conte	0.602
Most loving dog ever!	t3_1je1p2p	0.6

High-impact posts

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
That doesn't sound like a dog..... nobody will kill you....	Animal	2644	55	neutral	—	11.91
Recent video of me and my dog of 17 years who died this morning. he will always ...	Animal	1350	77	positive	—	11.57

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
Thank you to these people for saving the dog ❤️🐶	Animal	2352	38	positive	—	11.43
Dog's First Prosthetic	Animal	1234	47	neutral	—	10.99
This elderly man taking care of his elderly dog.. ❤️🐶👴	Animal	1661	34	positive	—	10.97

8. Post-level analysis (sample)

Showing up to 30 posts with VADER label, LR label, and extracted risk terms.

Title	Author	Ensemble	VADER	LR	Risk terms
My Dog's Ear are permanent 🐶	Tatofrioo	neutral	neutral	neutral	—
A picture of my dog with a painting of my dog	NinjaPlzGLB	neutral	neutral	neutral	—
Vote for Starla as Cutest Dog! ✨💖	PUPPERONI-PUPPY	positive	positive	positive	—
These are my own little puppies who entered a pet contest!	PUPPERONI-PUPPY	positive	positive	positive	—
Silly dog	[deleted]	neutral	neutral	neutral	—
I don't know how I got so lucky with my dog!	Imua-One	positive	positive	positive	—
Humanity Still Breathes on Our Streets	WrongdoerCharming417	positive	positive	positive	—
Balto, the Sled Dog Who Braved a Blizzard to Save a Town	Ok_Astronaut_6043	positive	positive	positive	—
Dad's working will Molly the office dog takes a nap!	ChuckGallagher57	neutral	neutral	neutral	—
"When you pick up the wrong 'dog' and it absolutely loves it."	Ok_Astronaut_6043	positive	positive	positive	—
Dog motor 🐾	WyattPurp23	neutral	neutral	neutral	—
Celebrating my friend's dog's birthday — party hat, treats, and lots o...	chaclaban	positive	positive	positive	—
This sheep is a dear	hasanpaintedthat	positive	positive	positive	—

Title	Author	Ensemble	VADER	LR	Risk terms
Do you brush your pets' teeth? Cats, dogs.. ?	Jean0406Alix	neutral	neutral	neutral	—
I need help, my friend is always at work and isn't able to properly ta...	AszaX	positive	positive	positive	—
Most loving dog ever!	THE_MrWen	positive	positive	positive	—
Photographer session with dogs	SurGregoRy	positive	positive	positive	—
Thank you to these people for saving the dog ❤️🐶	cutepamela8	positive	positive	positive	—
mushrooms	luvtheSavior	neutral	negative	positive	—
Looks like we have a home invader (possum)	ss_tall_toby_yt	positive	positive	positive	—
CUTE CAT & DOG COMPILATION	BaldandCorrupted	positive	positive	positive	—
This elderly man taking care of his elderly dog.. ❤️🐶	cutepamela8	positive	positive	positive	—
Dog and calf play together	Bliz_zy	positive	positive	positive	—
Calling All Dog Lovers 🙏	needstochill510	positive	positive	positive	—
Do pigeons eat fish? (GORE WARNING!!!)	ozhu_thrissur_kaaran	neutral	neutral	neutral	—
i got to meet two capys + a prairie dog	Brilliant-Diet-2430	neutral	neutral	neutral	—
Dogs can understand words and do maths – but there's problem UK Ne...	Beratungsmarketing	neutral	negative	positive	—
Wild Baby Bunny found. Looking for care tips??	Pompaditty	positive	positive	positive	—

Title	Author	Ensemble	VADER	LR	Risk terms
Dog I used to know, Photographed on November 24th, 2022, at 8:26pm	Strange-Wrongdoer-61	neutral	neutral	neutral	—
Cat the GOAT	YoouVish	positive	positive	positive	—

9. NLP techniques demonstrated (rubric)

Minimum **3 basic** + **3 advanced** techniques — implementation mapping for your written report.

Basic techniques

- **Preprocessing** (tokenize, stem, lemmatize, URL removal)
- **Named entity recognition** (spaCy / NLTK)
- **Part-of-speech tagging**
- **TF-IDF + Logistic Regression classification**
- **Sentiment ensemble** (VADER + LR majority vote)
- **Embedding-based topic clustering** (KMeans)
- **Regex information extraction** (hashtags, mentions, risk lexicon)

Advanced techniques

- **Foundation LLM** (OpenAI-compatible briefing)
- **Retrieval-augmented generation** (top-k posts)
- **Prompt engineering** (structured analyst template)
- **Chain-of-thought prompting**
- **LLM-as-judge / automated evaluation**
- **Ensemble methods** (sentiment + multi-signal crisis)
- **Agentic ReAct-style orchestration**
- **Ethics & fairness disclaimers**

10. Agent orchestration & evaluation

Agent status: completed

ReAct-style trace

Step	Thought	Action	Observation
1	Need Reddit posts about the brand/topic.	collect_posts	Starting data collection.
2	Collected 46 posts.	collect_posts	Received 46 posts.
3	Raw text must be normalized for NLP.	preprocess	Running tokenizer pipeline.
4	Preprocessing complete.	preprocess	46 posts prepared.
5	Extract hashtags, mentions, risk keywords.	extract	Regex IE.
6	Run NER and POS for entity-aware monitoring.	ner_pos	spaCy/NLTK.
7	Classify sentiment with VADER + LR ensemble.	sentiment	Ensemble vote.
8	Cluster topics with embeddings.	topics	KMeans clustering.
9	Compute temporal trends.	trends	Daily aggregation.
10	Score crisis risk from sentiment + engagement + trends.	crisis	Crisis formula.
11	Retrieve relevant posts for grounded LLM briefing (RAG).	rag	Query=dog
12	Generate executive briefing with CoT LLM.	briefing	LLM or fallback.

Automated evaluation

Sentiment agreement (VADER vs LR): 82.6% — VADER and LR agree on 38/46 posts (83%). Use manually labelled gold set for formal accuracy in your report.

Briefing judge score: 70/100 (error_fallback)

LLM judge unavailable; use manual rubric in report.

11. Ethics, limitations & references

--- *Ethics notice: Analysis uses public Reddit text. Sentiment and crisis scores are algorithmic estimates, not legal or PR advice. Models may reflect language bias; human review is required before brand actions.*

- Public Reddit text only; no private user data.
- Crisis scores are decision-support estimates, not legal or PR advice.
- Sentiment models can misclassify sarcasm and domain-specific language.
- For quantitative evaluation on labelled data, extend `backend/evaluation.py`.
- Full NLP technique reference: `docs/NLP_TECHNIQUES.md`

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